
REPORT OF ANALYSIS - PHYSICS, PARTICLE & SURFACE ANALYSIS

Sample:	MRa408
Order title/Product:	Graphene / Raman microscopy
Project:	Graphene
Account:	Curia Germany GmbH
Customer:	Dr. Thorsten Busch
Order no.:	25-049538-9
Remarks:	Raman measurement on request of the customer

Method:

Raman spectroscopy/microscopy using Senterra I Raman microscope (Bruker).

Procedure:

Objective:	Raman spectroscopy
Field of Application:	Powder
Principle:	The substance is applied to a microscope slide and examined under the microscope. The objective is selected and focused to examine an exact location using Raman. At this position, several points are examined with the laser and a corresponding number of Raman spectra are recorded.
Reference:	N/A

Experimental Details and Result

Equipment:	Senterra I Raman microscope (Bruker)
Magnification:	20 x
Wavenumber range:	45 – 3700 cm ⁻¹
Wavenumber step size	3 – 5 cm ⁻¹
Co-addition:	100
Exposure time:	12000 µm
Evaluation:	Averaging over similar spectra

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Result:

Depending on the laser position on the sample, two distinct types of spectra were recorded. Some spectra exhibited a clearly distinguishable 2D band (spectrum A), while others did not show this feature (spectrum B). Similar spectra were consolidated by averaging. The analysis of the integrals was performed separately for both types of spectra.

Spectra	2D band	G band	D band	D/G ratio
Typ A	12,0	223,0	285,0	1,278
Typ B	n.d.	245,4	331,7	1,352

Spectra see Fig 1.

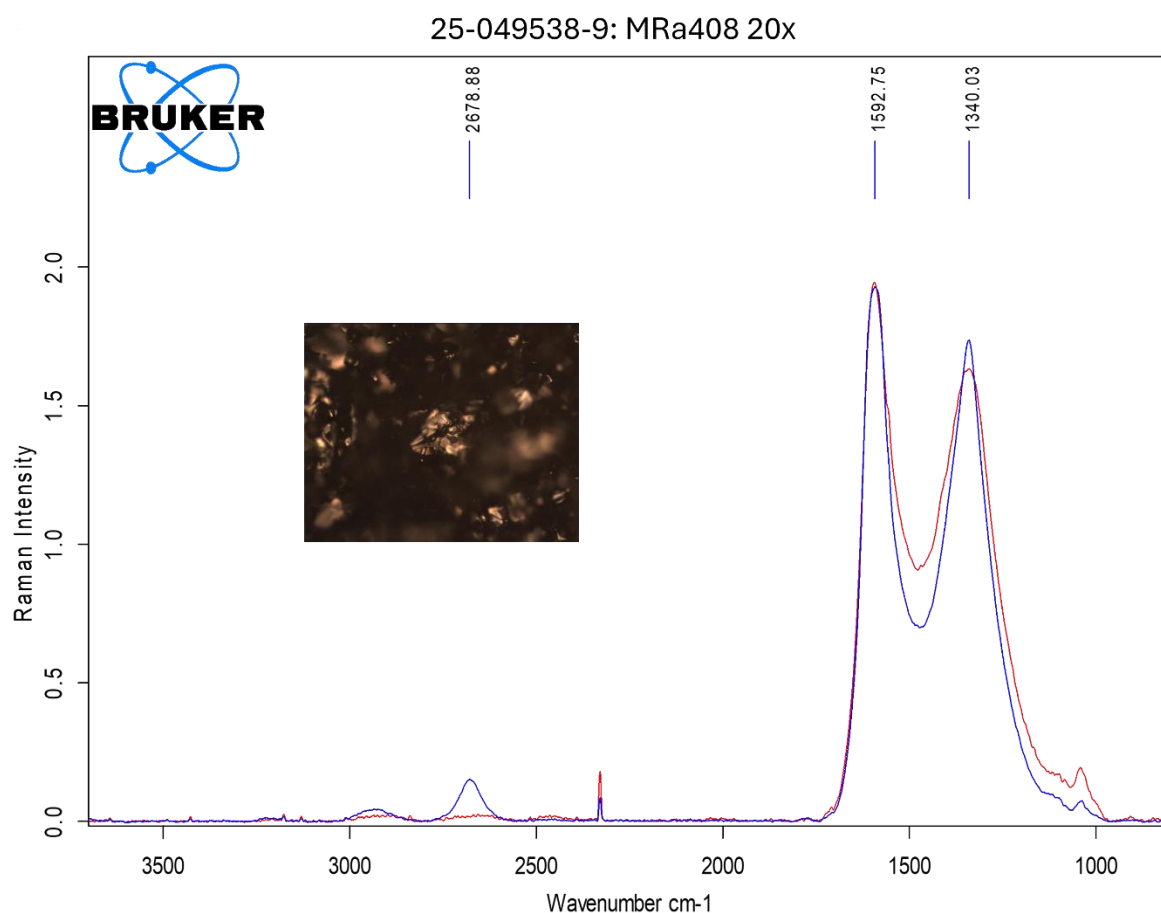


Figure 1: MRa408, blue curve: average spectrum over type A, red curve: average spectrum over type B

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